GEK-85710B



GE Drive Systems

INSTRUCTIONS

AF-200E™
ADJUSTABLE SPEED DRIVE
INSTALLATION, OPERATION AND MAINTENANCE

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes the matter should be referred to General Electric Company.

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SAFETY PRECAUTIONS

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment.

WARNINGS, CAUTIONS AND NOTES AS USED IN THIS PUBLICATION

WARNING

DENOTES OPERATING PROCEDURES AND PRACTICES THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CORRECTLY FOLLOWED.

CAUTION

DENOTES OPERATING PROCEDURES AND PRACTICES THAT, IF NOT STRICTLY OBSERVED, MAY RESULT IN DAMAGE TO, OR DESTRUCTION OF, THE EQUIPMENT.

NOTE

NOTES MERELY CALL ATTENTION TO INFORMATION THAT IS ESPECIALLY SIGNIFICANT IN UNDERSTANDING AND OPERATING THE EQUIPMENT.

WARNINGS

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLESHOOTING THE EQUIPMENT. FOLLOW THE START-UP PROCEDURE STEP BY STEP.
READ AND HEED ALL WARNING, CAUTION AND NOTE LABELS POSTED ON THE EQUIPMENT AND WITHIN THIS INSTRUCTION BOOK.
IMPROPER LIFTING CAN CAUSE SERIOUS OR FATAL INJURY. LIFT ONLY WITH ADEQUATE AND TRAINED PERSONNEL.

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. CIRCUIT BREAKERS, IF SUPPLIED AS PART OF THE TOTAL SYSTEM, MAY NOT DISCONNECT ALL POWER TO THE EQUIPMENT. SEE SYSTEM ELEMENTARY DIAGRAMS. WHETHER THE AC VOLTAGE SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGE TO GROUND MAY BE PRESENT AT MANY POINTS.

WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT GREAT CAUTION MUST BE USED. THE OSCILLOSCOPE'S CHASSIS SHOULD BE GROUNDED AND A DIFFERENTIAL AMPLIFIER INPUT SHOULD BE USED. CARE SHOULD BE USED IN THE SELECTION OF PROBES AND LEADS AND IN THE ADJUSTMENT OF THE OSCILLOSCOPE SO THAT ACCURATE READINGS MAY BE MADE. SEE OSCILLOSCOPE MANUFACTURER'S INSTRUCTION BOOK FOR PROPER OPERATION AND ADJUSTMENTS TO THE OSCILLOSCOPE.

SOME CONTROLLERS ARE FURNISHED WITH PARTIAL ENCLOSURES OPEN AT TOP AND BOTTOM. THESE ARE INTENDED ONLY FOR MOUNTING IN ANOTHER ENCLOSURE OR IN A CONTROL ROOM HAVING ACCESS BY QUALIFIED PERSONNEL ONLY.

EXPLOSIONS OR FIRES MIGHT RESULT FROM MOUNTING DRIVE CONTROLLERS IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE FLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. DRIVE CONTROLLERS SHOULD BE INSTALLED AWAY FROM HAZARDOUS AREAS, EVEN IF USED WITH MOTORS SUITABLE FOR USE IN THESE LOCATIONS.

ALL MOTOR BASES AND EQUIPMENT HOUSINGS SHOULD BE CONNECTED TO THE FACTORY OR FACILITY EARTH GROUNDING SYSTEM.

DRIVE MISOPERATION HAZARD: DO NOT APPLY ANY EXTERNAL REFERENCE VOLTAGE TO TERMINAL BOARD POINT TB-18 OTHER THAN VOLTAGE FROM THE MANUAL SPEED REFERENCE POTENTIOMETER. IF AN EXTERNAL REFERENCE IS TO BE USED, THE VOLTAGE FOLLOWER OPTION MUST BE INSTALLED AND ANY EXTERNAL REFERENCE VOLTAGE MUST BE WIRED TO THE TERMINAL BOARD ON THAT OPTION.

CAUTIONS

THE ELEMENTARY DIAGRAMS SUPPLIED WITH THE AF-200E CONTROL IDENTIFY SOME SIGNAL CONNECTIONS AS "LOW LEVEL ANALOG SIGNALS" AND INDICATE SPECIAL WIRING PRACTICES. THESE SIGNALS SHOULD BE ISOLATED FROM ALL OTHER WIRING (PARTICULARLY ANY POWER WIRES) AND RUN IN A SEPARATE CONDUIT OR WIRE RUN TO PREVENT POSSIBLE MISOPERATION OF THE CONTROLLER.

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HIPOT WITHOUT CONSULTING GE DRIVE SYSTEMS, GE COMPANY.

DO NOT CONNECT ANY EXTERNAL CIRCUITS OTHER THAN THOSE SHOWN ON THE ELEMENTARY DIAGRAM. CONNECTION OF DEVICES, OTHER THAN THOSE SHOWN, MAY DEGRADE THE PERFORMANCE OF THE DRIVE SYSTEM.

DO NOT USE POWER FACTOR CORRECTION CAPACITORS WITH THIS EQUIPMENT WITHOUT CONSULTING GE DRIVE SYSTEMS, GE COMPANY. DAMAGE MAY RESULT FROM HIGH VOLTAGES GENERATED WHEN CAPACITORS ARE SWITCHED.

NOTES

THE CONTROLLER COMMON (▽) CAN NOT BE GROUNDED. THIS MEANS THAT INPUT SIGNALS (i.e., PROCESS INSTRUMENT OUTPUT SIGNALS, ETC.) CAN NOT BE GROUNDED. IF A SEPARATE POWER TRANSFORMER IS USED AND IF THE SECONDARY OF THE TRANSFORMER MUST BE GROUNDED, IT IS RECOMMENDED THAT HIGH RESISTANCE GROUNDING BE USED.

1. INTRODUCTION

The AF-200E Adjustable Speed Drive control is a pulse width modulated inverter suitable for operating standard NEMA design B squirrel cage induction motors rated .5 to 5 HP with 208/230 VAC, 3-Phase, 60-Hz input or .5 to 10 HP with 460 VAC, 3-Phase, 60-Hz input. The controller converts a fixed frequency AC voltage input to a rectified potential DC voltage which is then inverted to an adjustable frequency, adjustable voltage power output. The speed of the motor is proportional to the output frequency which can be adjusted by either an externally mounted potentiometer or an isolated 4–20mA process instrument input signal with follower option.

Properly applied NEMA design B three-phase induction motors connected to the AF-200E control become adjustable speed motors with constant torque capability up to their base speed. Thus, fixed speed machinery can be converted to adjustable speed to optimize machine speed, process flow or otherwise improve productivity, efficiency and product quality. Some typical applications include:

- Fans
- Pumps
- Conveyors
- Textile Machines
- Woodworking Machines
- Machine Tools

2. RECEIVING, HANDLING AND STORAGE

2.1 RECEIVING

The equipment should be placed under adequate cover immediately upon receipt. Packing cases are not suitable for outdoor or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be obtained from **GE Company, GE Drive Systems, Erie, PA.** Wwhen seeking assistance, please use the drive serial serial number to identify the equipment. **Telephone: 814-455-0521**

2.2 HANDLING

Wall mounted controllers can be transported by lift trucks with the forks completely under the base, care being taken that the controller does not tip.

WARNING

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY. LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

2.3 STORAGE

If the equipment is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures from −20°C (−4°F) to 55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If storage environment exceeds a 15°C (27°F) drop in temperature at 50 percent relative humidity, over a 4-hour period, a source of heat should be made available in the near vicinity of the controller. Higher humidities with smaller temperature changes will also cause condensation.

3. INSTALLATION AND APPLICATION GUIDELINES

3.1 LOCATION

The AF-200E drive controller is suitable for most factory areas where other industrial equipment is installed. It should be installed in well-ventilated areas with ambient temperatures ranging from 0°C (32°F) to 40°C (104°F) and relative humidities up to 90°. It should be recognized however, that since life expectancy of an electronic component decreases with increased ambient temperature, the reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20°C (68°F) and 30°C (87°F).

Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dirt, dust and foreign matter.
2. Vibration and shock.
3. Moisture and corrosive vapors.
4. Temperature excursions outside specified limits.
5. Caustic fumes.
6. Power line fluctuations.
7. Electromagnetic interference (electrical noise).

WARNING

EXPLOSION OR FIRES MAY RESULT FROM MOUNTING THE CONTROLLER IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE INFLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. AF-200E CONTROLLERS MUST BE INSTALLED AWAY FROM HAZARDOUS AREAS EVEN IF USED WITH AC MOTORS SUITABLE FOR USE IN SUCH LOCATIONS.

3.2 MOUNTING

The controller enclosure may be mounted on any firm reasonably flat, vertical surface.

WARNING

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY. LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

WARNING

DO NOT SERVICE THE EQUIPMENT WHEN POWER IS APPLIED SINCE THIS COULD RESULT IN ELECTRICAL SHOCK OR BURN.

CAUTION

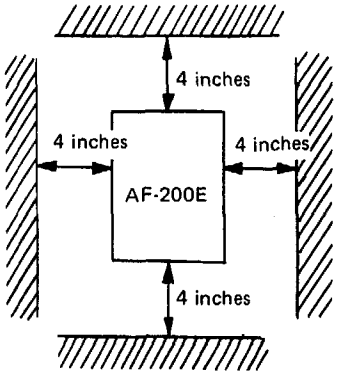
CARE SHOULD BE TAKEN TO SEE THAT ALL INTERCONNECTION WIRING IS SIZED AND INSTALLED IN ACCORDANCE WITH THE LATEST EDITION OF THE NATIONAL ELECTRICAL CODE (NEC) OR THE CANADIAN ELECTRICAL CODE (CEC) AND BE CONSISTENT WITH ALL OTHER APPLICABLE LOCAL CODES.

CAUTION

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HI POT WITHOUT CONSULTING DRIVE SYSTEMS, GE COMPANY.

3.3 CLEARANCE

This unit should be mounted with the following clearances:



3.4 LINE DISCONNECT AND THERMAL OVERLOAD RELAY

These must be supplied by the customer in order to meet the requirements of the National Electric Code. The line disconnect should be selected from items in Paragraph 4.6. The recommended thermal overload relay is a General Electric panel mount, manual reset only CR324, with heaters selected based on the applied motor's full load current at rated voltage.

3.5 AC LINE FUSES

Line fuses are recommended for protection of the unit. A list of recommended fuse sizes is shown in Table 14. These fuses are not provided with the drive and must be ordered separately.

3.6 AC SUPPLY REACTOR

A 3-Phase reactor must be installed if the AC supply impedance is too low relative to the controller rating. This external reactor is required to limit available short circuit current, to limit the DC link ripple voltage to within the capability of controller components, and to hold peak charging currents to acceptable levels.

Supply reactors should be installed when the capacity of the supply transformer is greater than ten (10) times the controller rating. (Refer to Table 12 for reactor selection.)

EXAMPLE:

Supply Transformer – 75 kVA
Inverter Rating – 10 HP
NO Reactor Required

4. APPLICATION NOTES

4.1 INVERTER UNIT

4.1.1 Due to the potential of damaging the unit, do not install capacitors for power factor improvement on the output of the inverter unit.

4.1.2 If a contactor is required between the inverter unit and the AC motor, the contactor should be opened or closed only when the AC motor is at rest. If the contactor is opened or closed during operation of the motor, the inverter may fault trip (over current) or damage components.

4.1.3 Motors with HP significantly greater than that of the inverter rating cannot be applied even if the motor load is light, because of its no load current.

4.1.4 The current limit level of the inverter unit is fixed at approximately 150% of its rated current. Therefore, the starting torque of the motor is 120% – 130% of the motor rated torque.

4.1.5 The inverter unit's and AC motor's braking torque is approximately 20% of the motor's rated torque. If the machine needs more braking torque, an optional dynamic braking kit (DB KIT) is available. (If DB KIT is required, refer to Paragraph 4.4 for Power Disconnect (for needed protection if a DB circuit failure should occur.)

4.1.6 Standard induction motors have approximately 3% speed regulation from no load to full load, due to motor slip.

4.2 AC MOTOR

4.2.1 Standard AC motors must be derated for constant torque loads applied over a wide speed range to prevent overheating at low speeds, where the motor cooling fan is not very effective. Derating is usually not necessary for fan or pump type loads, where load torque decreases with decreasing speed.

4.2.2 Standard AC motors must be checked for overspeed capability with the manufacturer when they are used at high speed (more than 60Hz) in order to avoid mechanical problems.

4.2.3 When motors with mechanical brakes are used, the power supply for the brake should be taken from the input side of the inverter. The brake timing should be synchronized with the inverter operation. (See Paragraph 7.5.)

5. AF-200E COMPONENT DESCRIPTION

5.1 MAIN PRINTED CIRCUIT BOARD (C200020AL G03, G04)

This card contains the necessary regulating transistor firing and signal processing circuitry to control the drive. All drive adjustment jumpers and potentiometers are located on this card. (See Section 9, ADJUSTMENTS.) A set of diagnostic and drive status LEDs is also located on this card.

5.2 RECTIFIER MODULE

The input rectifier module converts the 3-Phase AC input voltage to a fixed DC voltage or "link" which is filtered by the link filter capacitor. Depending on controller rating, the rectifier module may be a single three-phase module (lower horsepower ratings) or may be three single-phase modules connected in a three-phase rectifier bridge.

5.3 FILTER CAPACITOR

The filter capacitors reduce the ripple from the rectifier module and provide a smooth DC voltage which is applied to the inverting power transistors.

5.4 CHARGE CONTROL CIRCUIT

The charge control circuit is made up of a resistor which is connected between the rectifier module and filter capacitors and a charge control relay which shorts out the resistor. The resistor is used to limit the inrush charging current from the rectifier to the filter capacitors; the relay shorts out the resistor after the initial inrush current has subsided.

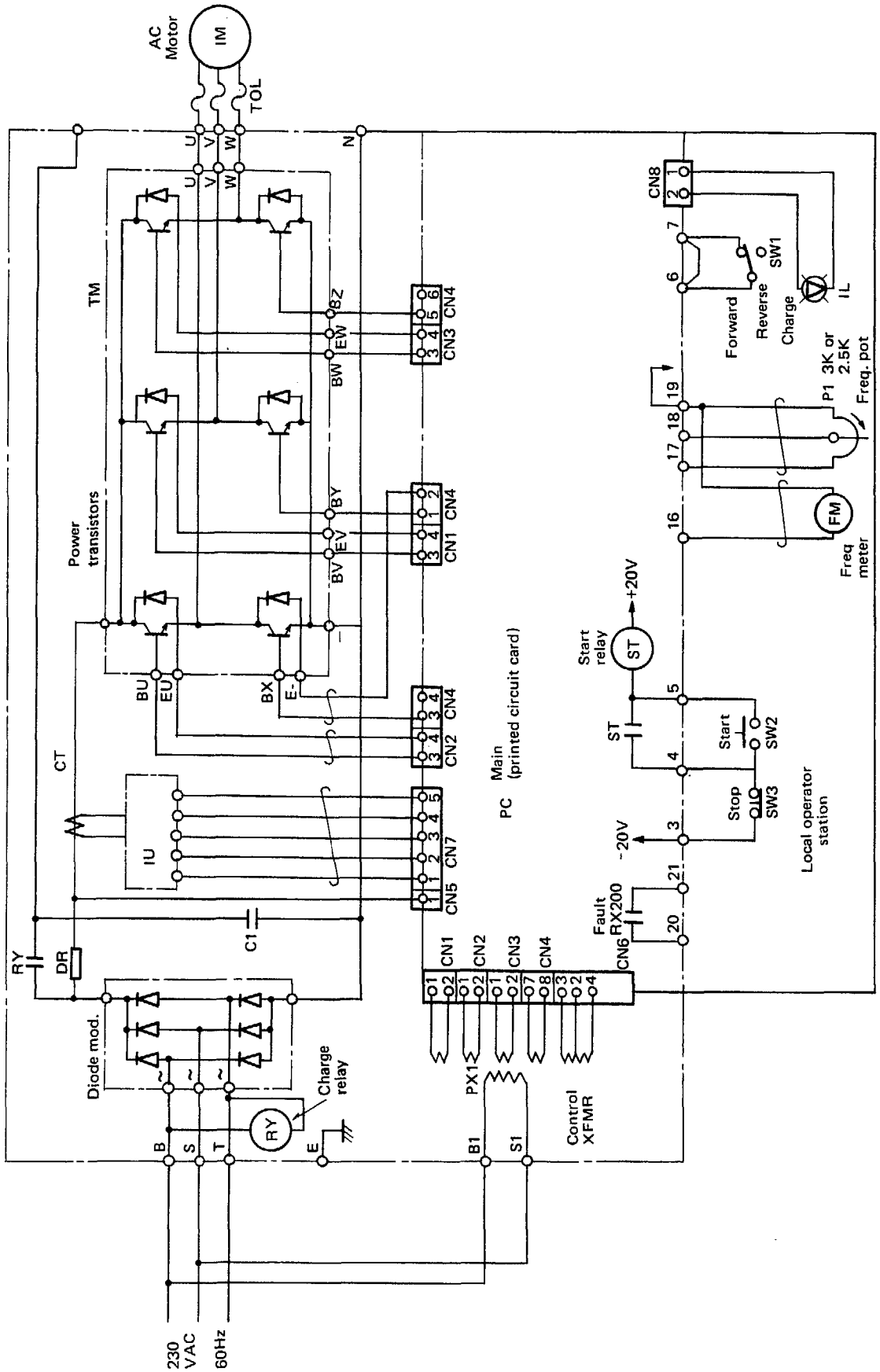
5.5 CONTROL POWER TRANSFORMER

The control power transformer provides the special isolated power supply voltages required by the Main Printed Circuit Board.

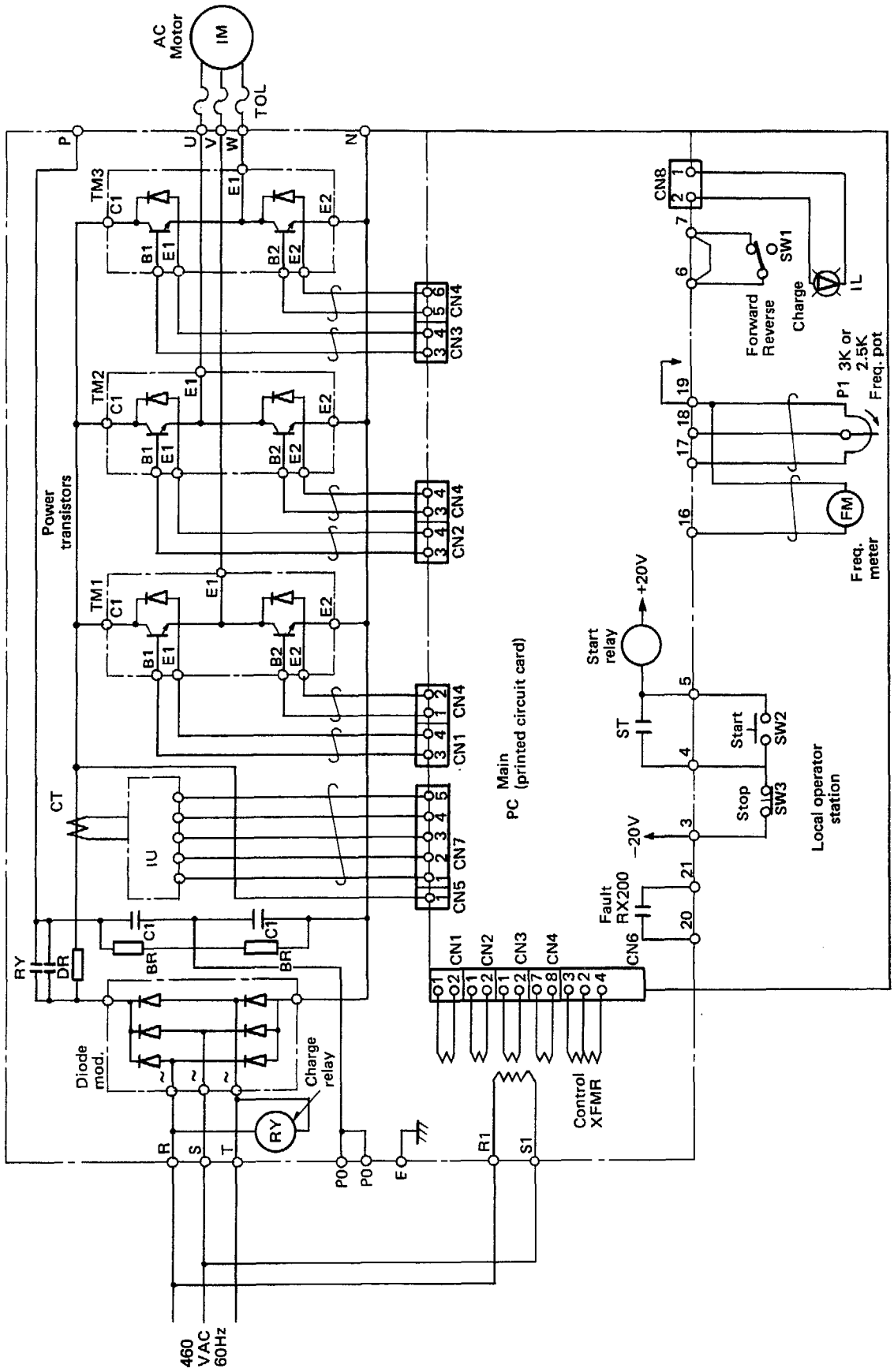
5.6 INVERTER POWER TRANSISTORS

The power transistors are used to switch the filtered DC link voltage on and off in such a manner as to create a near-sinusoidal, three-phase output wave shape. By controlling the switching frequency of these devices, an adjustable output voltage/frequency can be applied to a standard induction motor, thus varying its speed.

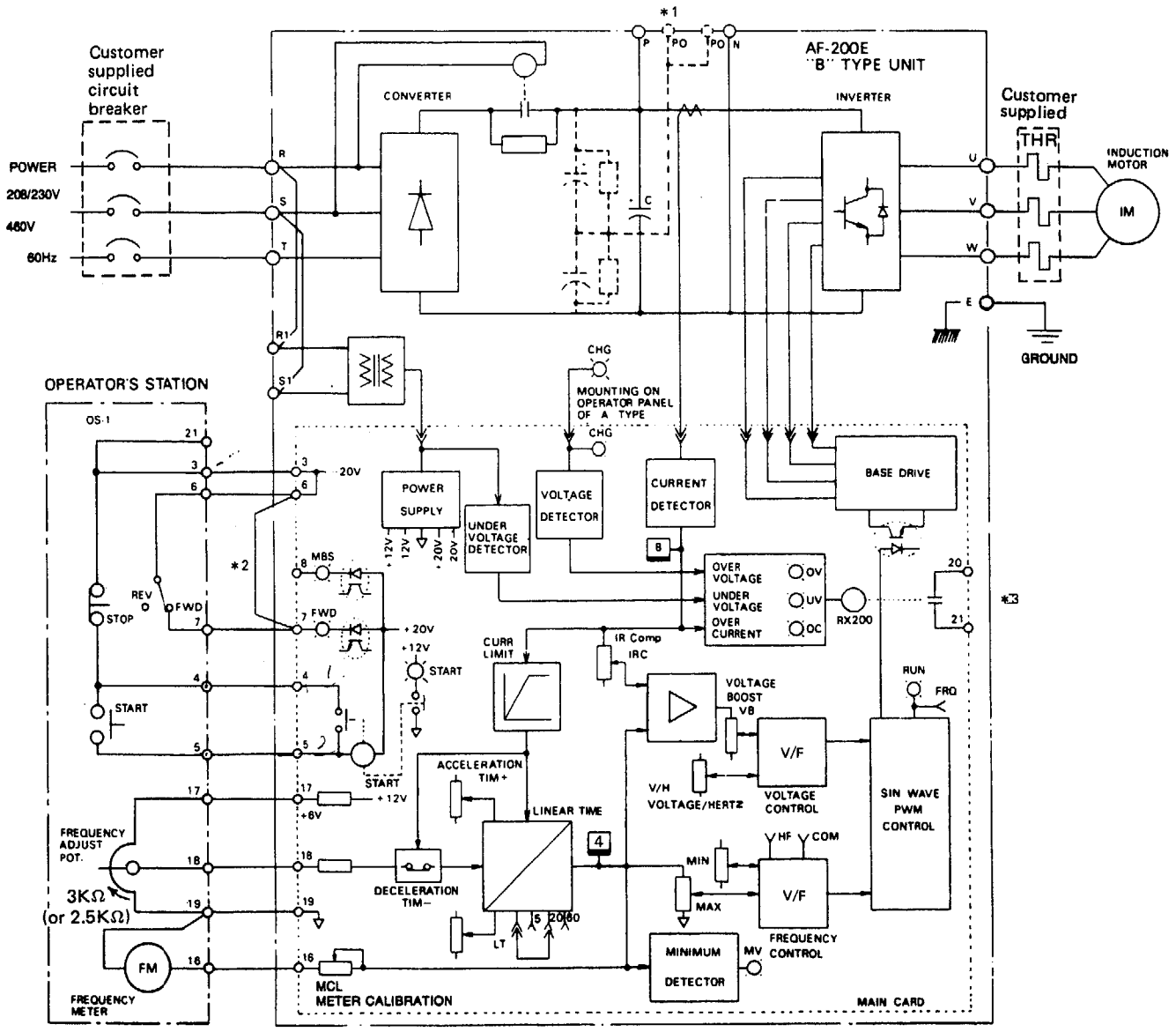
5.7 TYPICAL POWER AND CONTROL ELEMENTARY 230 VAC UNIT



5.8 TYPICAL POWER AND CONTROL ELEMENTARY 460 VAC UNIT



6. BLOCK DIAGRAM



Notes

*1 Use only 460V class unit with DB KIT option (See paragraph 13.3)

*2 For Rev. operation, remove jumper 6-7.

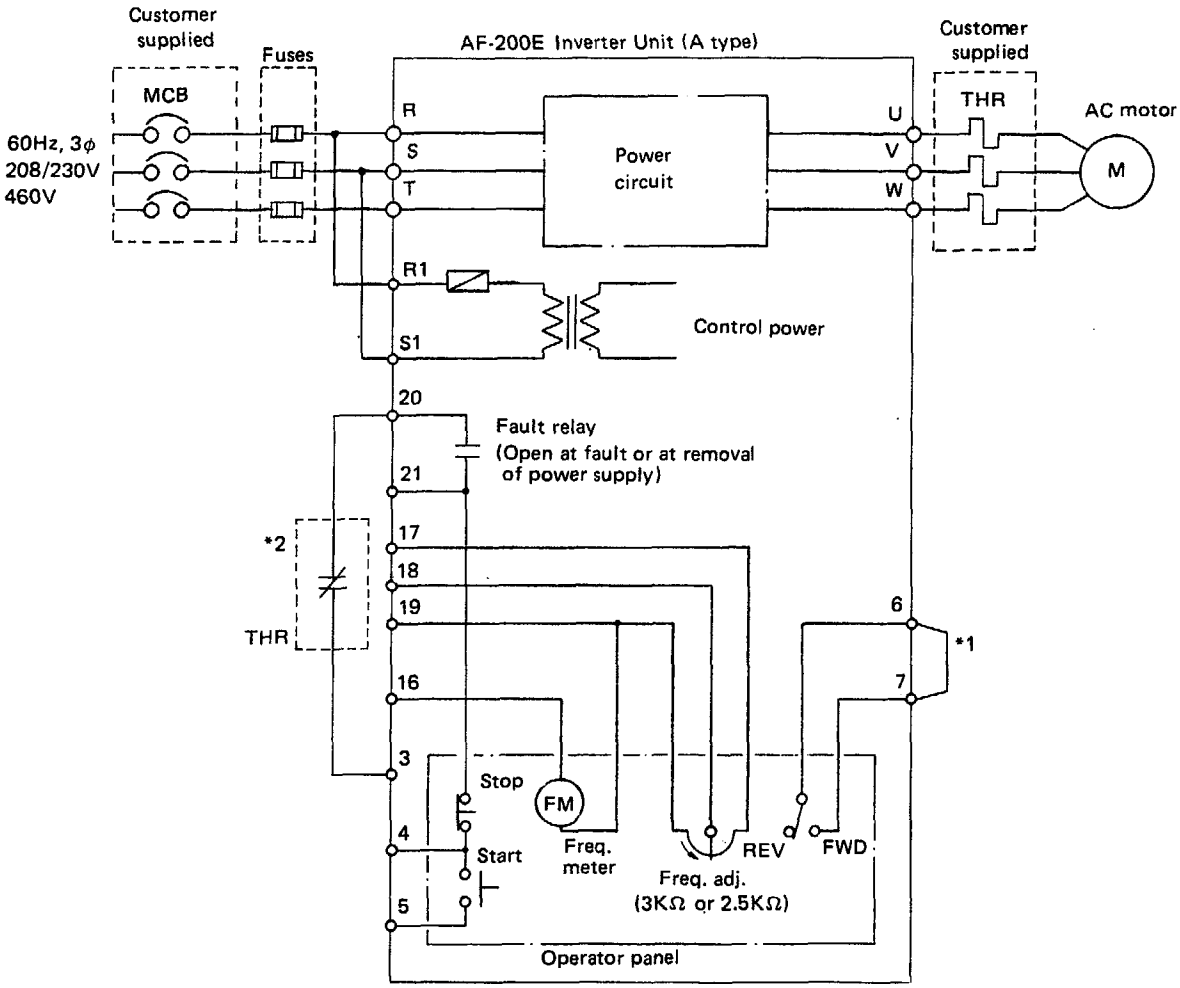
*3 "Inverter unit trip" signal for external use. Refer to paragraphs 7.1 thru 7.5 for proper use. Open at fault or at removal of power supply.

*4 ⊡ : Test point

7. WIRING

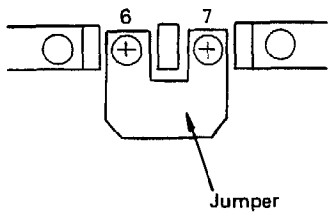
Following are typical connection diagrams.

7.1 WITH LOCAL OPERATOR PANEL (A TYPE)



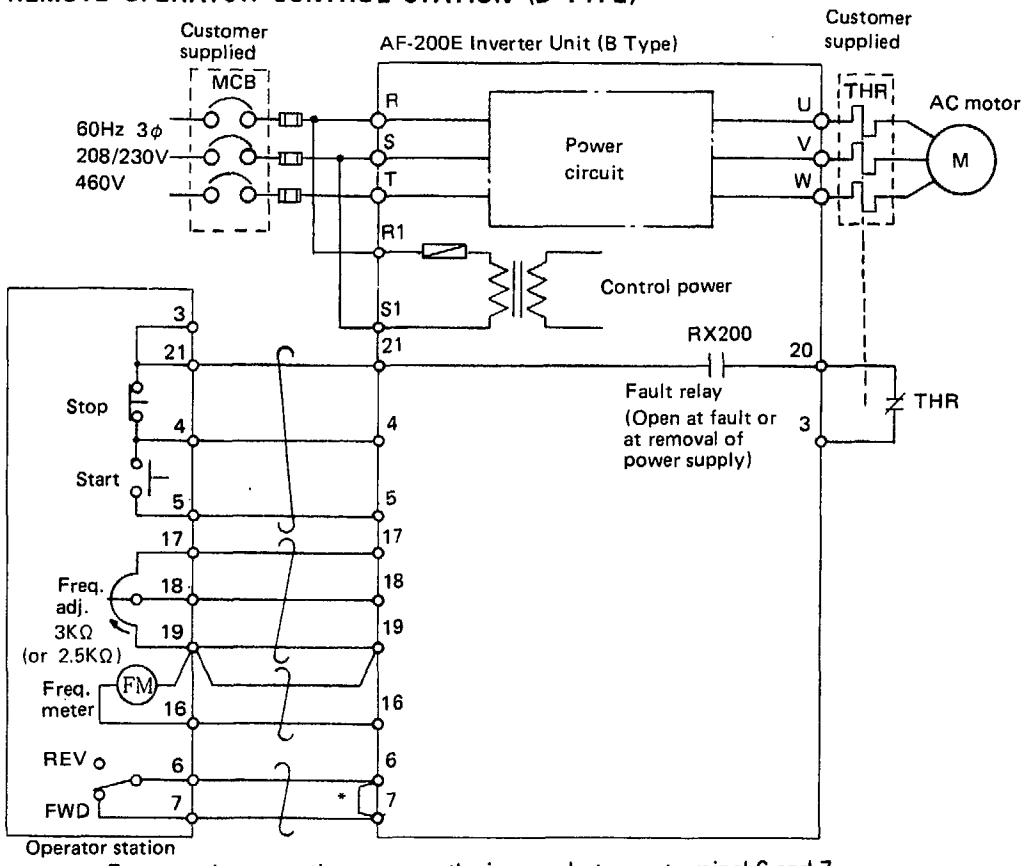
Note

*1. For reversing operation, remove the jumper between terminal 6 and 7.



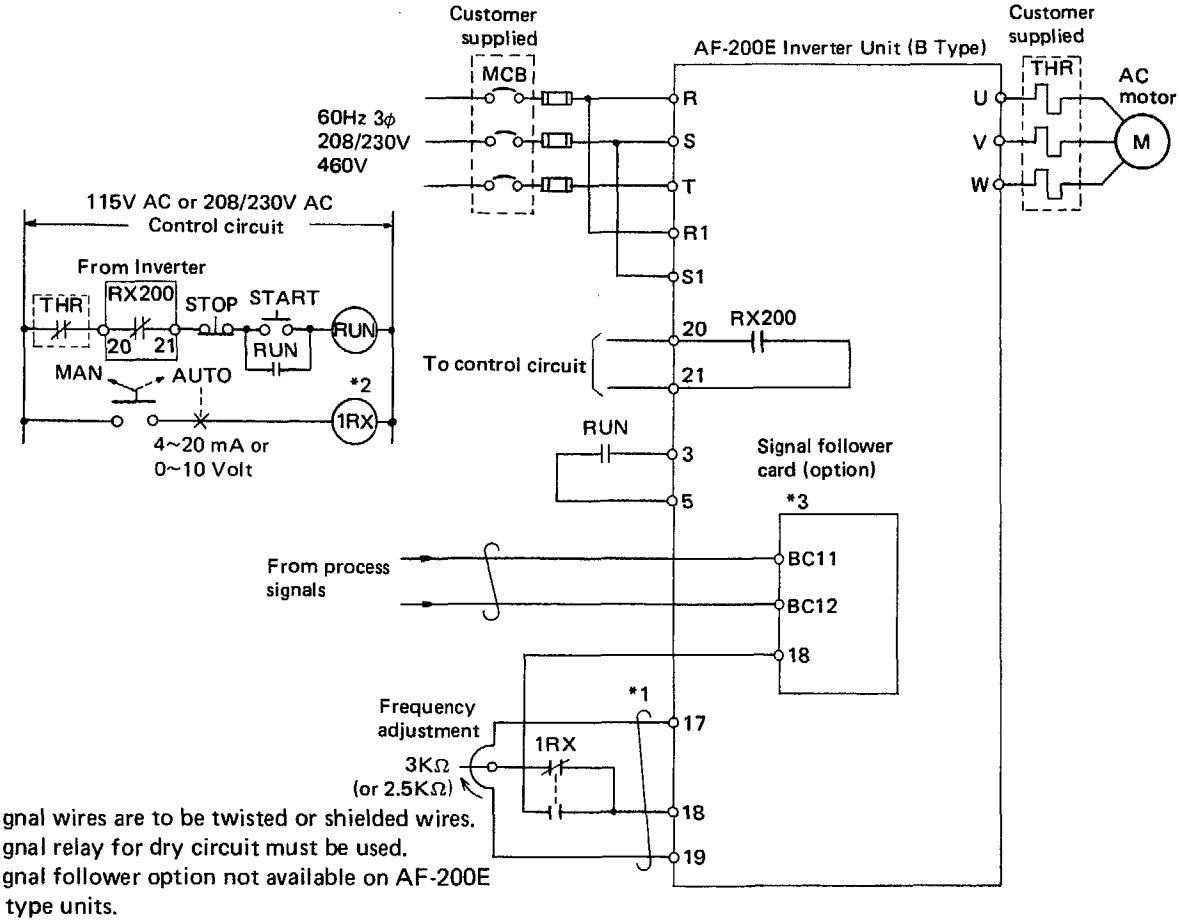
*2. Remove the jumper between terminal 3 and 20. Connect thermal overload contact supplied by customer.

7.2 WITH REMOTE OPERATOR CONTROL STATION (B TYPE)



* : For reversing operation, remove the jumper between terminal 6 and 7.

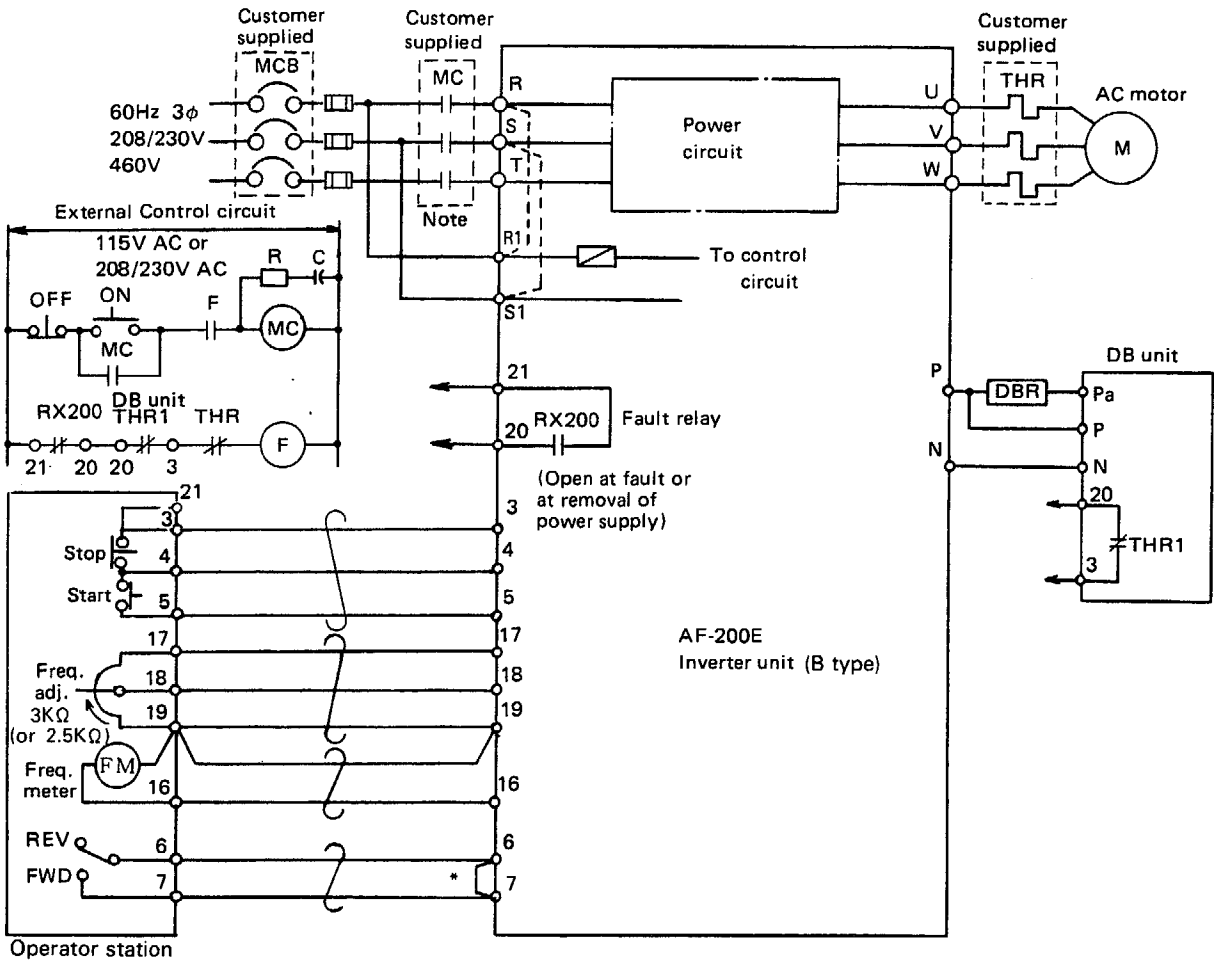
7.3 WITH REMOTE OPERATOR'S CONTROL AND SIGNAL FOLLOWER CARD (B TYPE)



Notes

*1. Signal wires are to be twisted or shielded wires.
*2. Signal relay for dry circuit must be used.
*3. Signal follower option not available on AF-200E A type units.

7.4 WITH REMOTE OPERATOR STATION, DB KIT AND CUSTOMER SUPPLIED LINE CONTRACTOR (B TYPE)



Note: MC is required for mechanical disconnection of DB unit in case of DB failure should occur.
* For reversing operation, remove the jumper between terminal 6 and 7.

7.5 WITH MECHANICAL BRAKE (B TYPE)

7.6 TERMINAL CONNECTIONS

The table below shows the power and control connections.

Table 1: Terminal Connections

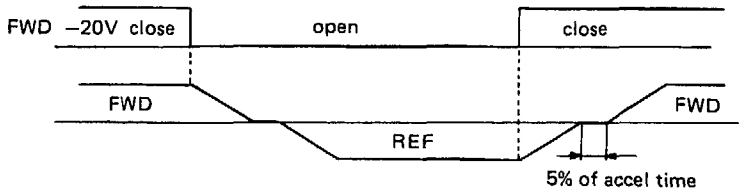
	Nomenclature	Terminal No.	Description
Power terminals *1	R, S. T		AC input power
	U, V, W		Output power
	R1, S1		AC input power for the control power transformer
	P, N (PO, PO)		Positive and negative terminals of the DC link
	E		Earth terminal
Control terminals (on printed circuit board) *2	FRQ	16	A frequency meter connection Max. 2mA DC output
	+REF	17	Speed reference potentiometer connection
	REF	18	Refer to the connection diagram (7.1 to 7.5)
	COM	19	Common connection
*3	START 2	5	The start signal connection
	START 1	4	Refer to the connection diagram (4-1 to 4-4)
	FWD	7	Reverse operation signal connection. Refer to the connection diagram (7.1 to 7.5)
*4	MBS	8	Mechanical braking signal connection. When this terminal is connected to −20V(6), the inverter output is forced off and motor coasts to a stop.
	−20V	3, 6	Minus 20VDC terminal
	Remove	20, 21	Fault relay contact. Open at fault or at removal of power supply. Contact rating 250V AC max 3A resistive, 0.5A inductive.

Notes

*1 For 460VAC input, two "PO" terminals are added for two sets of DB units and DB resistors. See Section 6 "Block Diagram".

*2 If an input signal, such as (4 ~ 20mA) is required, a follower card (option) for interfacing is available, refer to Paragraph 13.4, PROCESS FOLLOWER KIT.

*3 When TB7 and 8 are opened the motor decelerates to a stop and accelerates in the reverse direction to set speed.



*4 MBS is used for operating contactors on the inverter output when motors with magnetic brakes are used.

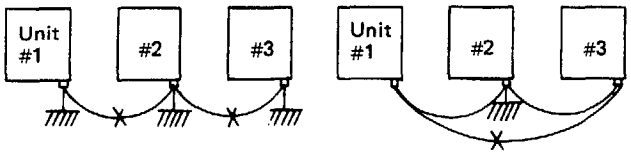
7.7 WIRING

7.7.1 Wire sizes for power lines and main circuit breaker ratings can be selected in accordance with the following table.

Table 2: Wiring

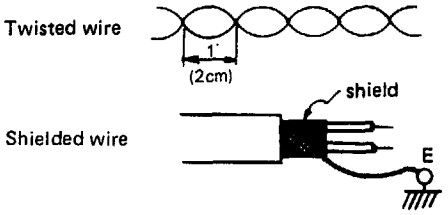
AC input	AC motor		Power wire size		Main circuit breaker
	HP	Rated current	(AWG No.)	(mm^2)	
208/230V	0.5	2.6 A	#14	2.0	TEB132010WL (10A)
	1	4.2	#14	2.0	
	2	7.5	#14	2.0	TEB132015WL (15A)
	3	10.5	#12	3.5	TEB132020WL (20A)
	5	16.0	#12	3.5	TEB132030WL (30A)
460V	0.5	1.8	#14	2.0	TED134010WL (10A)
	1		#14	2.0	TED134010WL (10A)
	2	3.3	#14	2.0	TED134010WL (10A)
	3	4.7	#14	2.0	TED134010WL (10A)
	5	7.9	#14	2.0	TED134015WL (15A)
	7.5	11.4	#12	3.5	TED134020WL (20A)
	10	14.5	#12	3.5	TED134030WL (30A)

7.7.2 "E" terminal on the inverter unit should be grounded by AWG#12 wire.
If several units are to be grounded, grounding wires should be connected as follows. (Either scheme is acceptable.)



7.7.3 The control signal interconnecting wires must be run in magnetic conduit separate from power wires to eliminate the possibility of electrical noise pick up.
Shielding wires or twisted wires should be used for signal interconnecting in parallel. Inside the panel the control signal wires must be twisted, and must not run with power wires.
If control signal wires must be run in parallel with power wires, shielded wire should be used.

7.7.4 If shielded wires are used, each wire should be grounded at inverter end only. The following are methods recommended for twisting wires and connecting shields.



7.7.5 Nearby relays, solenoids or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added across the coil of these devices. A 220 ohms, 2 watts resistor in series with a 0.5 mfd, 600 volts capacitor should be used in 100VAC and 200/220VAC circuits, and 460V circuits should use 0.5 mfd capacitor 1000V or higher.

8. START-UP

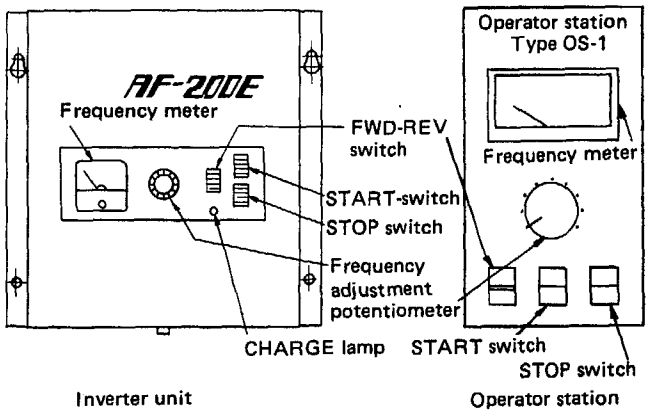
8.1 INITIAL INSPECTION

8.1.1 With power OFF, check that all connections are right.

8.1.2 Check that the supply voltage is 208/230V or 460V ±10% AC.

8.1.3 The AC motor voltage, current and frequency nameplate rating must agree with the controller rating.

8.2 START-UP PROCEDURE



8.2.1 Set the operator's frequency adjustment potentiometer to zero (completely counter clockwise).

8.2.2 Close the disconnect switch to apply AC power to the inverter unit. The charge lamp on the unit and the MV lamp on the card should turn on.

8.2.3 Press the operator's start pushbutton, (or close the RUN switch) and observe the green RUN lamp on the card. It should be blinking at a frequency of 2 to 3 hertz, and the start lamp on the card should turn on.

8.2.4 Gradually increase the operator's frequency adjustment potentiometer. The MV lamp should turn off, the motor should begin to rotate and the green RUN lamp should increase frequency.
If the motor rotation is opposite, stop the motor. Disconnect AC power, and reverse the connection to any two motor leads. Restart the inverter unit.

8.2.5 Gradually increase the speed potentiometer up to the maximum speed setting while observing motor current and motor operation. The motor should run smoothly up to its rated speed and the current of the connected motor should remain fairly constant over its entire speed range.

8.2.6 Press the operator's STOP push button. The motor speed should decelerate smoothly down to a slow speed. The RUN lamp frequency should decrease. The RUN lamp should then go out and the motor should stop.

Note: If the inverter unit trips out for any reason disconnect the AC power to reset the unit.

8.3 INDICATING LIGHTS ON THE CARD

Operation or fault conditions will be annunciated by lights on the main control card.

Table 3: Indicating Lights

Nomenclature	Description
RUN	Operating frequency indication
START	Operation mode
FWD	Forward rotating mode
MBS	Forced to turn off mode
CHG	Charged condition of the DC link capacitors
MV	No reference voltage or current limit
UV	Under voltage fault
OC	Over current fault
OV	Over voltage fault

9. ADJUSTMENTS

The potentiometers on the card are adjusted before shipping.
If readjustment is needed, refer to the following table.

9.1 Table 4: Potentiometers

Nomenclature	Description	By turning CW	Factory setting
TIM+	Accel. time adjustment	Increase the time (Note 1)	3 sec. "LT-20" jumper
TIM-	Decel. time adjustment	ditto	ditto
V/H	V/Hz adjustment	(Note 2)	(Note 2)
MAX.	Max. frequency adjustment	Increases the frequency (Note 3)	60Hz (120Hz available)
MIN.	Min. frequency adjustment	Increases min frequency (Note 3)	4Hz
VB	Voltage boost adjustment	Increases the output voltage (Note 5)	27V/6Hz (54V/6Hz)
IRC	IR-Comp. adjustment	ditto (Note 4)	MAX CCW
MCL	Meter calibration	Increases the indicating value	60Hz

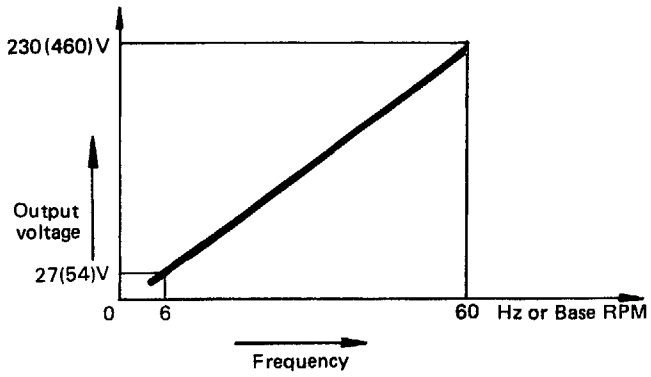
Notes
1) Accel. and decel. time adjustment range can be selected by a jumper on the printed circuit board.
(See "AF-200E card components layout", Section 14.)

9.2 Table 5: Linear Time Jumpers

Acceleration or Deceleration Time	Jumper selection
0.5 ~ 6.0 seconds	"LT"—"6" pins on the card
3 ~ 20 seconds	"LT"—"20" pins on the card
7 ~ 60 seconds	"LT"—"60" pins on the card

If the setting of time of "TIM+" or "TIM-" is too fast with large inertia machines an overcurrent (OC) or overvoltage trip may occur. Set the time longer.

2) The unit is factory set at 210V 60Hz for 208/230V input, 420V 60Hz for 460V input. An increase in V/H adjustment is not ordinarily required for most standard AC motors.



3) This adjustment allows setting of minimum inverter starting frequency to optimize starting torque (depending on motor characteristics). Clockwise rotation increases minimum frequency. Normally set at 1.5 times the motor slip freq.

4) Operation for more than 60Hz to 120Hz.
If greater than 60Hz operation is required readjust per the following procedure.

a) Jumper "HF"–"COM" on the main control card by the jumper which is attached to the backside of the AF-200E front panel.

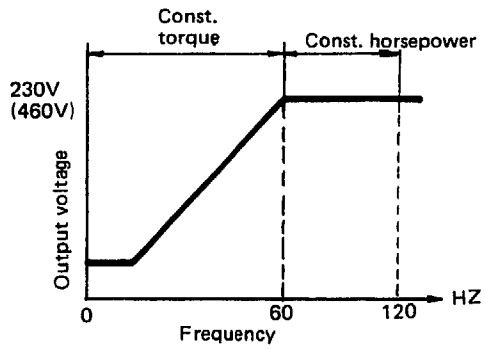
b) Connect a frequency counter across "FRQ"–"COM" pins for observing frequency, or use a hand tach on the motor to check RPM.

c) Start the inverter and gradually turn potentiometer clockwise to max., but not to exceed 120 Hz.

d) Adjust "MAX" pot on the main control card to get desired frequency.

e) Adjust "MCL" pot so that frequency meter indicates desired frequency.

Note; Inverter output voltage should not exceed rated AC input voltage.



5) IRC increases the output voltage at low speed to compensate the voltage drop in proportion to motor load. This compensated voltage becomes zero at more than 40% of rated frequency.
IRC is factory adjusted, however, if motor speed is unstable at low speed and heavy load operation, turn IRC clockwise until the motor becomes stable.

6) VB (Voltage boost)
This adjustment is used to change (either increase or decrease) the amount of torque the motor will produce at speeds below 1/4 (one-fourth) rated. It may also be used to increase the amount of breakway torque the motor can produce (within the motor's design limitations). VB should be adjusted such that the motor(s) accelerate smoothly from rest; otherwise, the adjustment should remain fully CCW.

7) Potentiometer VCP. This potentiometer must be set full CCW or output voltage surging may occur.

10. TROUBLESHOOTING

10.1 TYPES OF METERS

The following types of meters are recommended for troubleshooting.

Measurement	Type of meter
Input AC line voltage and current	Moving iron type. ( )
Output voltage	Rectifier type. ( )
Output current	Moving iron type. ( )

10.2 TEST TABS ON THE MAIN CONTROL CARD

Three test tabs are provided for troubleshooting.

Table 6: Test Points

Tab number or name	Description	Normal meter reading
4 and COM	Output of the linear time circuit	0 ~ –6VDC
8 and COM	Current through DC link	+2VDC at 100% rated current
FRQ and COM	Operating frequency	0 +10V

10.3 FAULT INDICATING LIGHTS

If any of the fault indicating lights are lit, it may be due to the following causes.

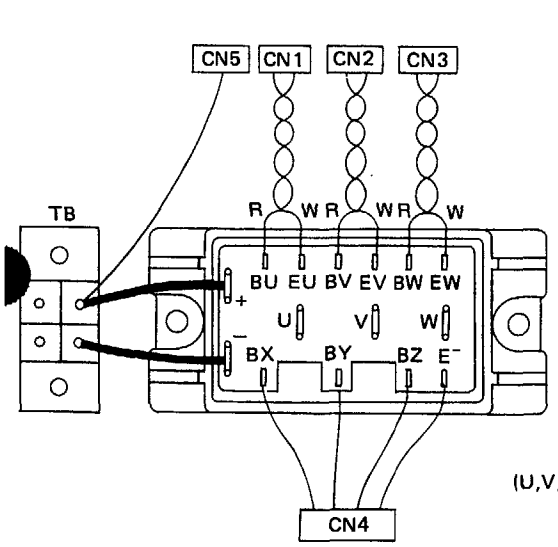
Table 7: Fault Ligts

Fault lamp	Possible causes
OC (over current)	a) Too short acceleration time b) Overload or impact loading c) Restarted into rotating motor d) Short circuit or grounding of the output
OV (over voltage)	a) Too short deceleration time b) Too high input line voltage c) Overhauling load
UV (under voltage)	a) Input line voltage is too low. b) Power loss (more than 15 mili second)

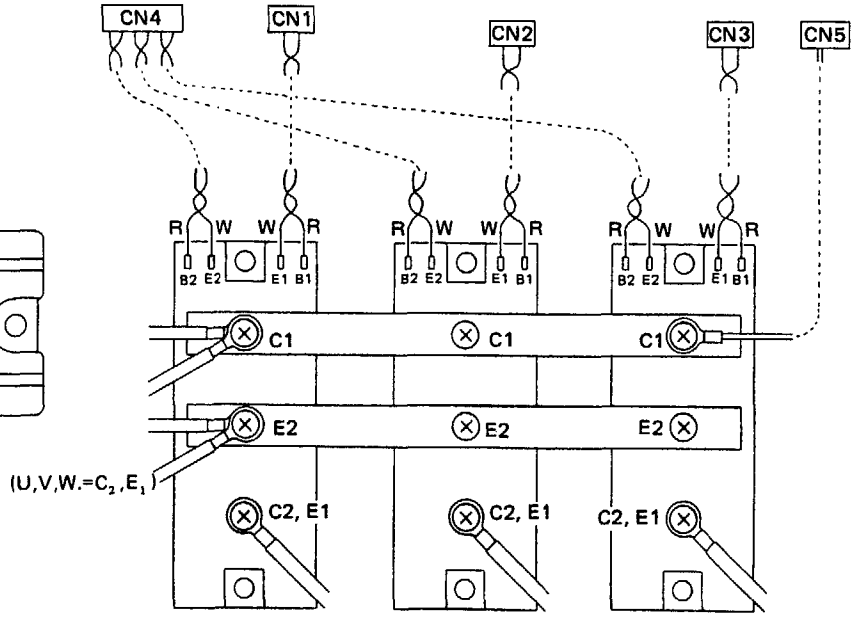
10.4 CHECKING OF THE POWER TRANSISTOR MODULES

If a faulty power transistor module is suspected, the module can be checked by a multi-meter, by disconnecting CN1–CN5 connectors on the main control card and removing bus bars on the power modules.

Connect meter lead to		Meter reading for good modules
Positive lead	Negative lead	
B 1 B 2	C 1 C 2	~~0 to 500~~
E 1 E 2	C 1 C 2	200 – 1000Ω ~~0 to 500~~
B 1 B 2	E 1 E 2	0 to 500
E 1 E 2	B 1 B 2	0 to 500
C 1 C 2	B 1 B 2	50K to ∞

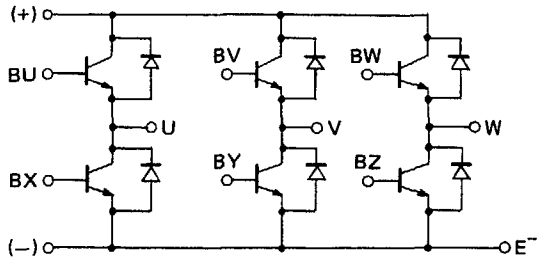


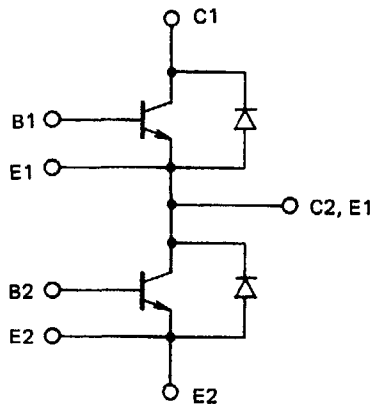
For AC208/230V 0.5/1 HP



For AC208/230V 2 ~ 5HP
AC460V 0.5 ~ 10HP

Layout of the power transistor modules.

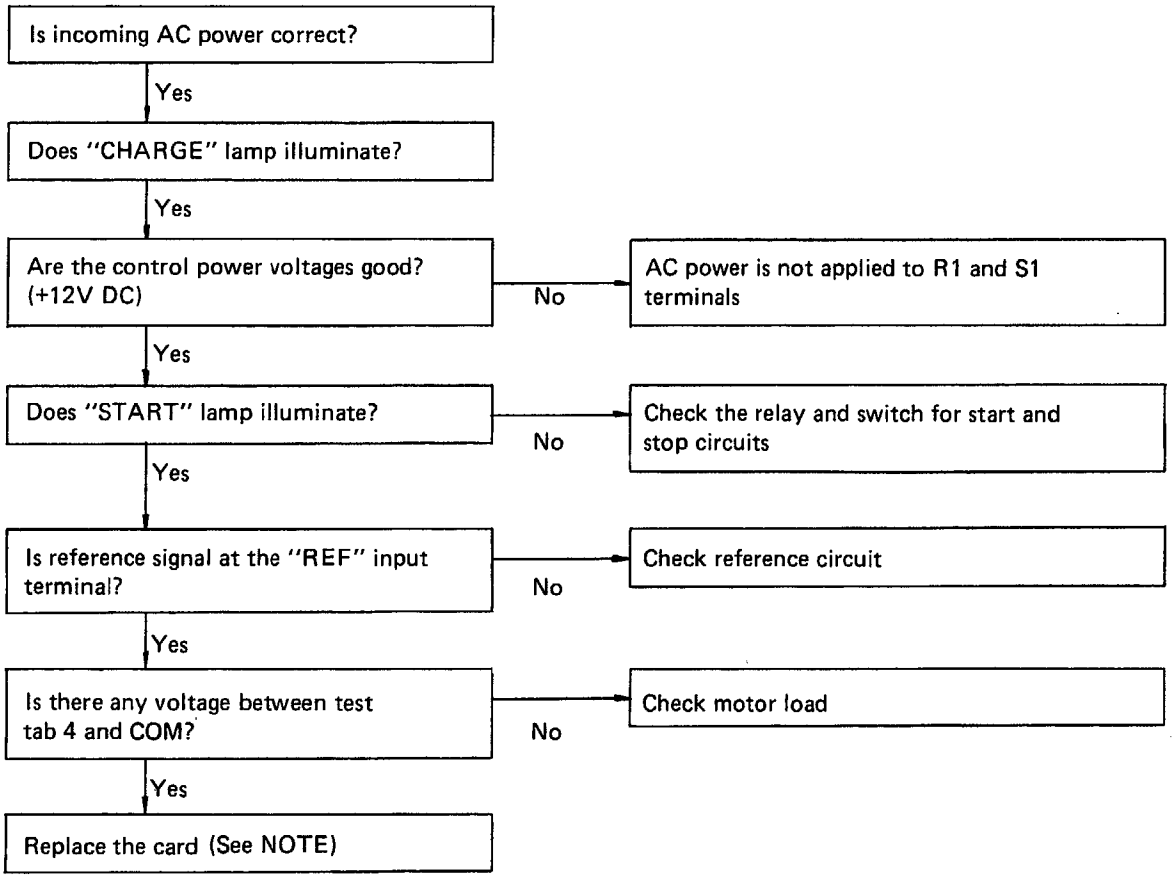




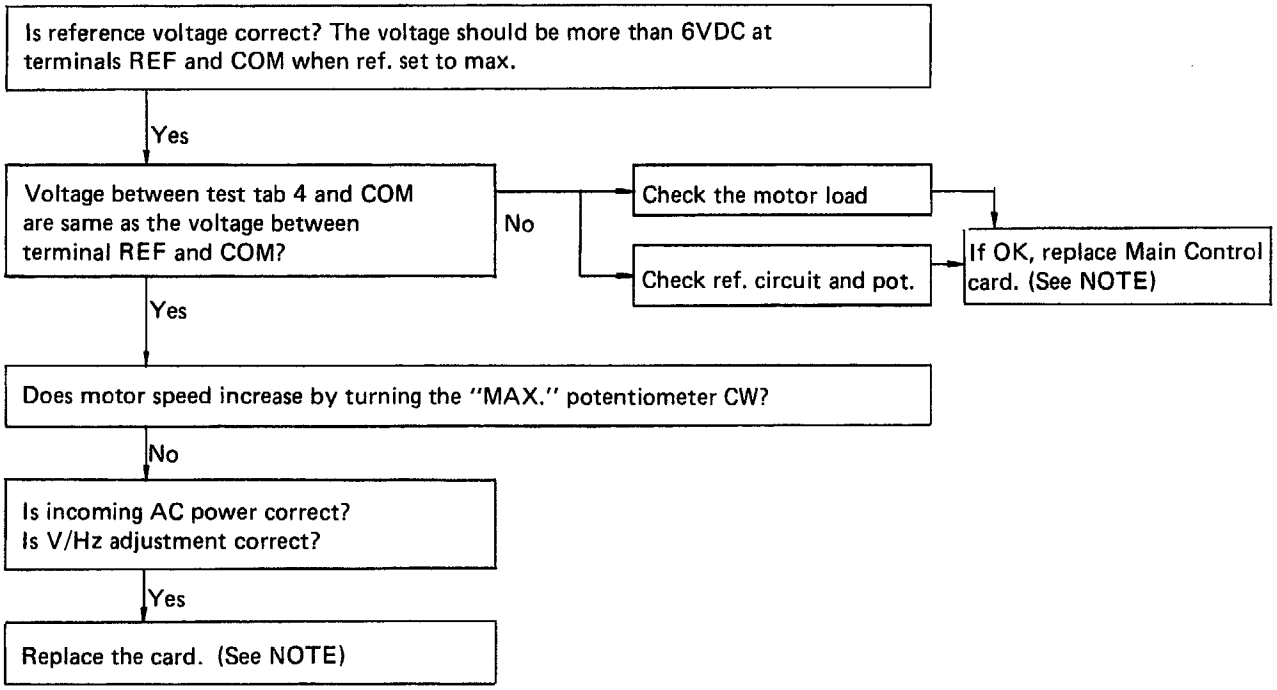
Equivalent circuit of the power transistor modules.

10.5 TROUBLESHOOTING CHART

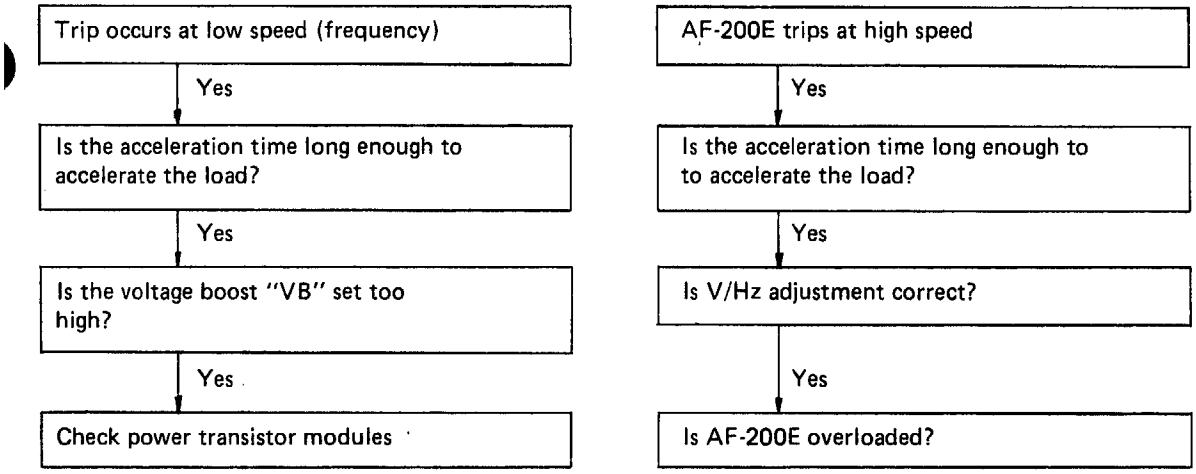
A. Motor does not run.



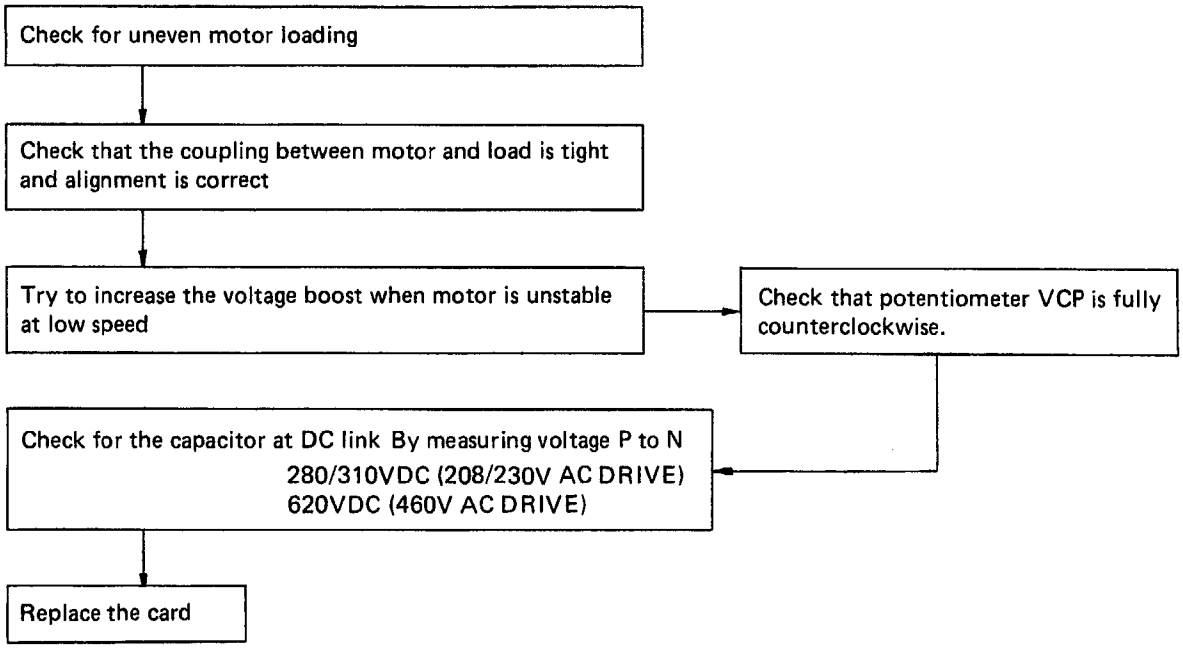
B. Cannot obtain rated frequency or speed.



C. Can start but trips when speed reference is increased.



D. Motor speed is unstable or not smooth



NOTE: Main Printed Circuit Board Replacement – If replacing the Main Printed Circuit Board is necessary, place all jumpers on the replacement board in the same position as on the failed board. Also set all potentiometers on the replacement board to the same setting as on the failed board. Make sure potentiometer VCP is fully CCW.

11. SPARE AND RENEWAL PARTS

Renewal parts are essential to avoid costly downtime and maintain your equipment. Recommended spare parts are as follows.

11.1 AC INPUT VOLTAGE 208/230V 60HZ

Table 8: 208/230 VAC Drives

Name	CAT. No.	Specification	0.5/1HP	2HP	3HP	5HP
			QTY			
Transistor module	104Y170AA 066	15A	1			
	104Y170AA 040	30A		3	3	
	104Y170AA 041	50A				3
Diode module	101X301AA 032	10A	1	1		
	101X301AA 026	15A			1	
	101X301AA 027	30A				1
Capacitor	104Y122AA 272	680MF	1	1		
	104Y122AA 270	1500MF			1	
	104Y122AA 271	2200MF				1
Main Card	C200020AL** G03	For 208/230V	1	1	1	1

** ; May be changed for improvement.

11.2 AC INPUT VOLTAGE 460V, 60HZ

Table 9: 460 VAC Drives

Name	CAT. No.	Specification	0.5/1HP	2HP	3HP	5HP	7.5HP	10HP
			QTY					
Transistor module	104X170CB 010 (104Y170AA 070)	50A	3	3	3	3	3	3
Diode module	101X301AA 046	50A	1	1	1	1		
	101X301AA 045	100A					1	1
Capacitor	104Y122AA 272	680MF	2	2	2			
	104Y122AA 270	1500MF				2		
	104Y122AA 271	2200MF					2	2
Main Card	C200020AL** G04	For 460V	1	1	1	1	1	1

** ; May be changed for improvement.

12. STANDARD SPECIFICATIONS & MODIFICATIONS

12.1 STANDARD SPECIFICATIONS

Table 10: Standard Specifications

	Series	AF-200E series 208/230 V class				AF-200E series 460 V class					
	Application Motor (HP) Note 1	0.5/1.0	2.0	3.0	5.0	0.5/1.0	2.0	3.0	5.0	7.5	10.0
Output	with operator panel	6VAF251A	6VAF252A	6VAF253A	6VAF255A	6VAF261A	6VAF262A	6VAF263A	6VAF265A	6VAF266A	6VAF267A
	without operator panel	6VAF251B	6VAF252B	6VAF253B	6VAF255B	6VAF261B	6VAF262B	6VAF263B	6VAF265B	6VAF266B	6VAF267B
	Rated output current (A)	4.5	7.5	11.0	16.0	1.8	3.3	4.7	7.9	11.4	14.5
	Rated output KVA	1.6/1.8	2.7/3.0	4.0/4.3	5.7/6.3	1.4	2.6	3.7	6.3	9.0	11.5
Weight (lbs)		17.6	17.6	18.3	18.3	19.4	20.3	20.3	20.3	20.3	20.3
Construction		Enclosed & wall mounted									
Power Supply	Voltage frequency	3-phase, 208/230 VAC 60 Hz				3-phase, 460 VAC 60 Hz					
	Permissible voltage regulation	Within ±10%				Within ±10%					
	Permissible frequency regulation	Within ±5%									
Enviromental Conditions	Ambient temperature	0 to +40° C									
	Ambient humidity	90% relative humidity non-condensing									
	Altitude	Sea level to 3300 feet									
	Atmosphere	To be free from corrosive gas and dust									
	Vibration	Less than 0.5G									
Control Specifications	Control method	Sinusoidal wave pwm control									
	Frequency range	1:10 (6-60 Hz constant torque, 100 Hz or 120 Hz constant HP)									
	Output voltage Note 2	3-phase AC									
	Frequency accuracy	Within ±0.5% at max. frequency									
	Voltage/Frequency ratio	Constant V/Hz									
	Overcurrent	150% of rated current for one minute									
	Frequency reference signal	DC 0 to +6V (input impedance 30K ohms)									
	Acceleration/Deceleration time	STD (3 - 20 sec.) with jumper selection, (0.5 - 60 sec. available)									
	Regenerative braking torque	Approx. 20%, (100% braking torque with DB brake option)									
Protection Functions	Undervoltage protection	If input voltage is 15% of rated input voltage or under, the transistor circuits are shut off and protected. An "UV" indication lamp is provided.									
	Prevention of stall caused by regenerative overvoltage	If regenerative braking torque exceeds 20% of rated level while decelerating, further decrease of frequency is stopped until torque is restored to the normal level. Under severe conditions the transistor circuits are shut off and protected. An "OV" indication lamp is provided.									
	Overcurrent protection	The overcurrent protection circuit will operate and the transistor circuits will be shut off when an overcurrent larger than 200% of rated output current occurs. An "OC" indication lamp is provided.									
	Instantaneous power failure	Protection circuits will operate and the transistor circuit will be shut off, if the power failure exceeds 15 milliseconds.									

Note 1: The applicable motor output in the table assumes a General Electric NEMA design B induction motor used.
Note 2: Output voltage cannot be more than input voltage.

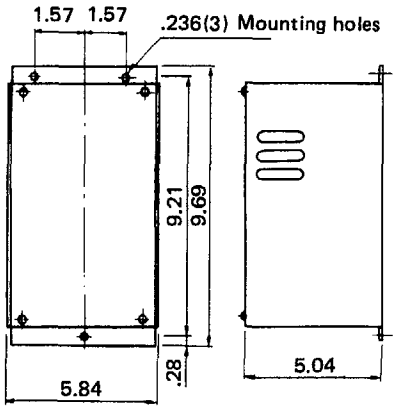
13. MODIFICATIONS

13.1	**REMOTE OPERATOR'S STATION 6VOC 23A1**	Consists of start/stop switch, foward/reverse selector switch, frequency setting potentiometer and frequency meter. (Mounted remote to inverter unit by customer.)
13.2	**SIGNAL FOLLOWER CARD 6VPF20KA1 FOR 4–20mA 6VVF10KA1 FOR 0–10V**	For external signal, 4-20mA, or 0-10V DC. (Cannot be used with local operator's station)

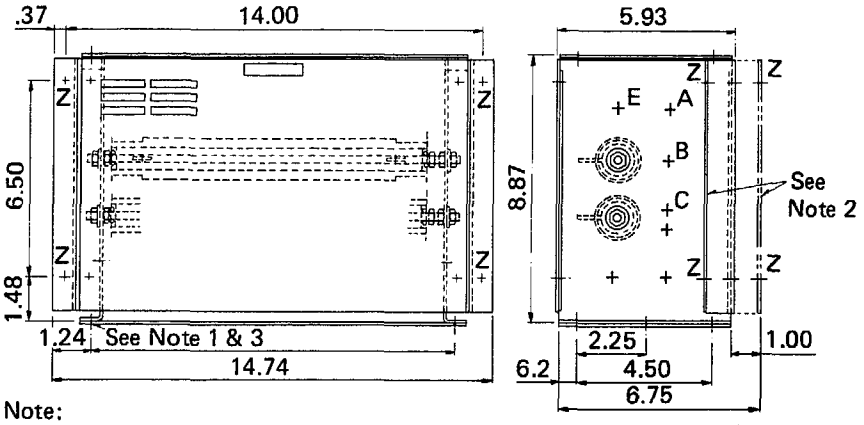
13.3 DYNAMIC BRAKING (DB) KIT (OPTIONAL)

13.3.1 DB Unit (DBU–0A)

Dimensions in inches



13.3.2 DB Resistor Enclosure



Note:

1. View shown is for bottom MTG with (6) .218 x .437 slots.
2. For wall or back MTG. remove back plate and remount so it projects the (4) .218 x .437 Z MTG slots to be against the MTG surface.
3. Bottom cover to be removed if bottom mounting is used.

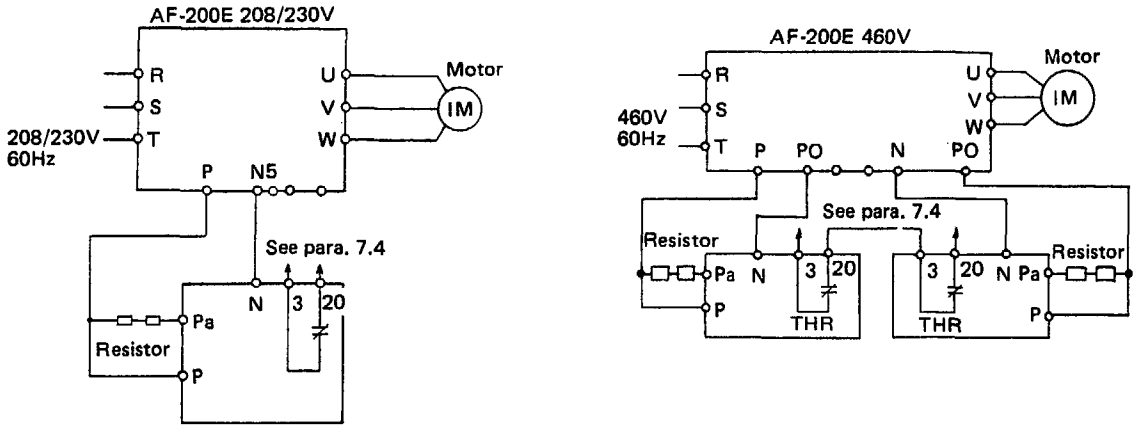
13.3.3 DB Kit (DB Unit + Resistor)

Table 11: Wire Size Parameters

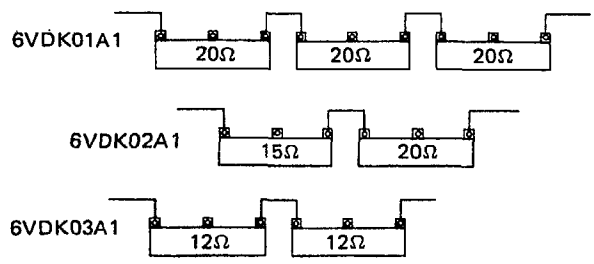
100% BRAKING TORQUE 30 SEC.

RATING / CAT No.	AC208/230V				AC460V						Wire Size	
	1HP	2HP	3HP	5HP	1HP	2HP	3HP	5HP	7.5HP	10HP	P, N, Po	3, 20
6VDK01A1	1				2	2					#14 (2mm²)	#14 (2mm²)
6VDK02A1		1					2	2			#14 (2mm²)	#14 (2mm²)
6VDK03A1			1	1					2	2	#12 (3.5mm²)	#14 (2mm²)

13.3.4 DB Kit Wiring



Resistor wiring



Notes

1. Wiring except as shown above for DB KIT is same as shown in Paragraphs 7.1 thru 7.5.
2. For 460V application, 2 set of DB KITs are required.
3. Refer to Section 4 for protective devices.

DB KIT and WIRING

13.4 PROCESS FOLLOWER KIT (OPTION)

The signal follower card which is mounted on AF-200E B type inverter, makes it possible to operate the inverter by process signals for 4–20mA or 0 ~ +10V.

13.4.1 Connection

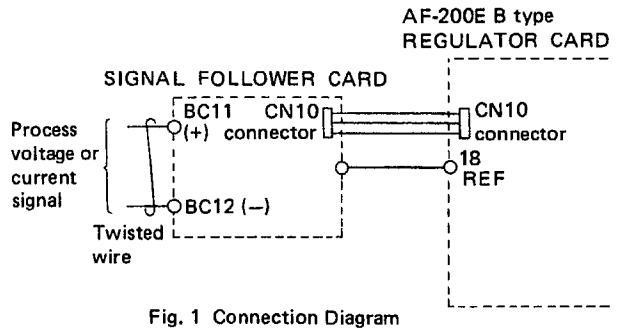


Fig. 1 Connection Diagram

Note 1: Connect positive signal to BC11 and negative to BC12.
Note 2: AF-200E A type (Inverter with built in operator panel) cannot be used with signal follower card.

13.4.2 Basic Theory

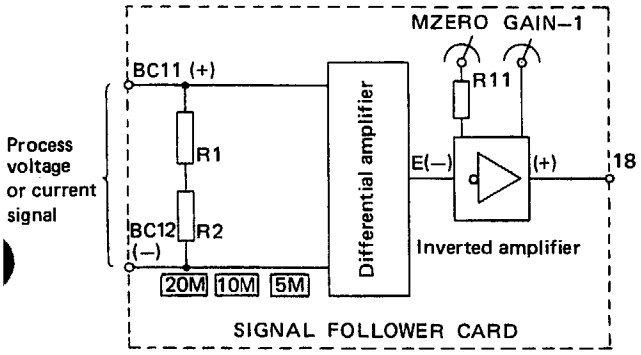


Fig. 2 Signal Follower Card Block Diagram

The process signal, (an external input signal), is fed into BC11 and BC12 terminals on the card, where the process current signal is converted into a voltage signal, which is amplified and inverted into a negative signal by the differential amplifier.

The output (E) of differential amplifier is trimmed at the inverting amplifier by "MZERO" and "GAIN–1" potentiometers. Then its positive output appears at terminal 18 of the card. This signal must be wired to terminal 18 on AF-200E inverter card.

If process signal is voltage, the differential amplifier acts as an inverting buffer and other instructions are the same as described in the current signal, except "MZERO" potentiometer, which is not used.

13.4.3 Group Classification

The KIT is classified into following groups depending on input signals.

	Input signal	Output of TB18	Cat No.
Current signal	4 ~ 20 mA	0 ~ 6V	6VPF20KA1
Voltage signal	DC0 ~ +10V	0 ~ 6V	6VVF10KA1

13.4.4 Adjustment

All adjustments have been made at the factory. However if a slight adjustment is required, use the following procedure;

○ Operation by process current signal

(a) Adjustment of minimum speed
Required min. motor speed is adjusted with potentiometer "MZERO" at min. input current as shown in Fig. 3.

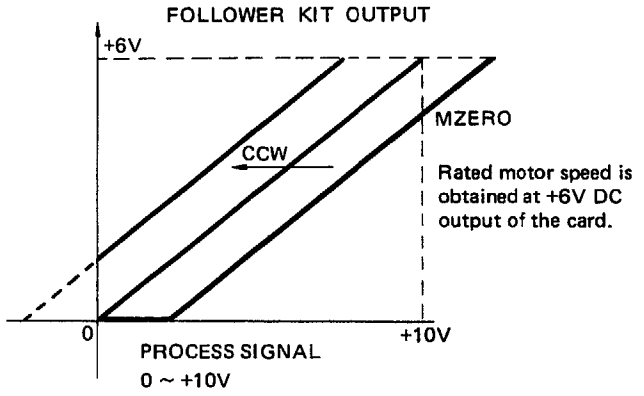


Fig. 3 Minimum speed adjustment

(b) Adjustment of maximum speed
Required max. motor speed is adjusted with potentiometer "GAIN–1" at max. input current, as shown in Fig. 4.

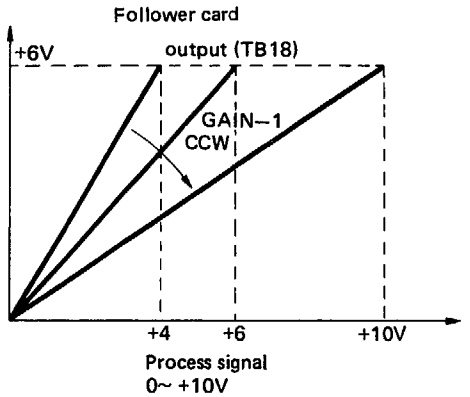
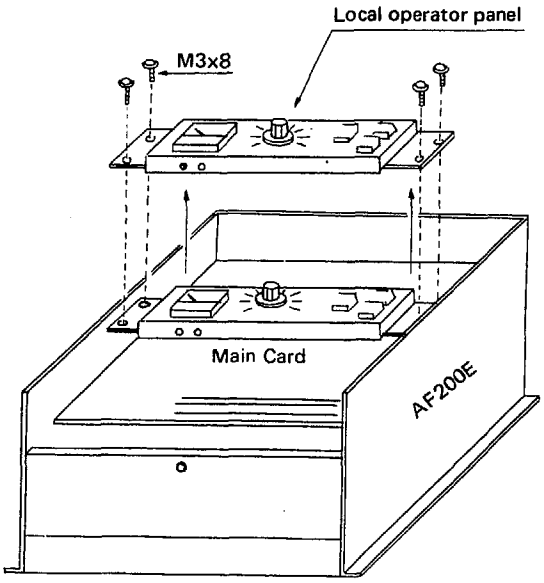


Fig. 4 Maximum speed adjustment

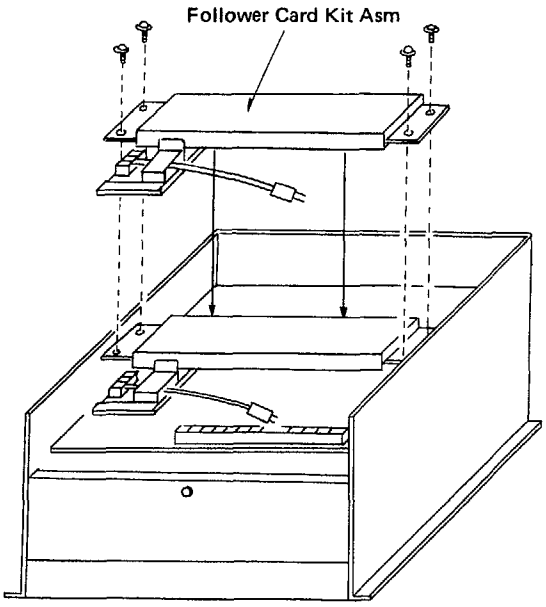
13.4.5 AF-200E Follower Card Kit Asm Mounting Procedure

CAT No. 6VPF20KA1 for 4-20mA input (ungrounded)
6VVF10KA1 for 0–10V input (ungrounded)

13.4.5.1 a) Remove 4 screws from local operator panel
b) Disconnect CN8 connector (2P) from main card.
c) Disconnect wires at TB 3,4,5,6,7,16,17,18 and 19 on main card.
d) Take out local operator panel.

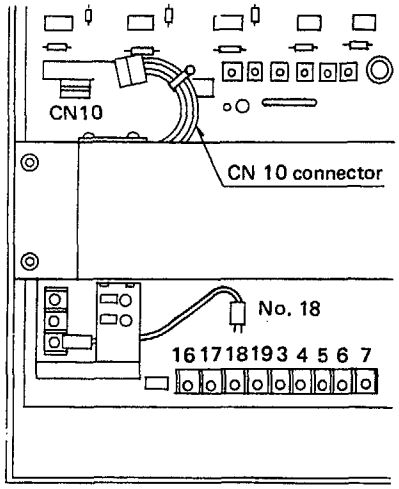


13.4.5.2



a) Secure follower card kit asm with 4 screws.

13.4.5.3

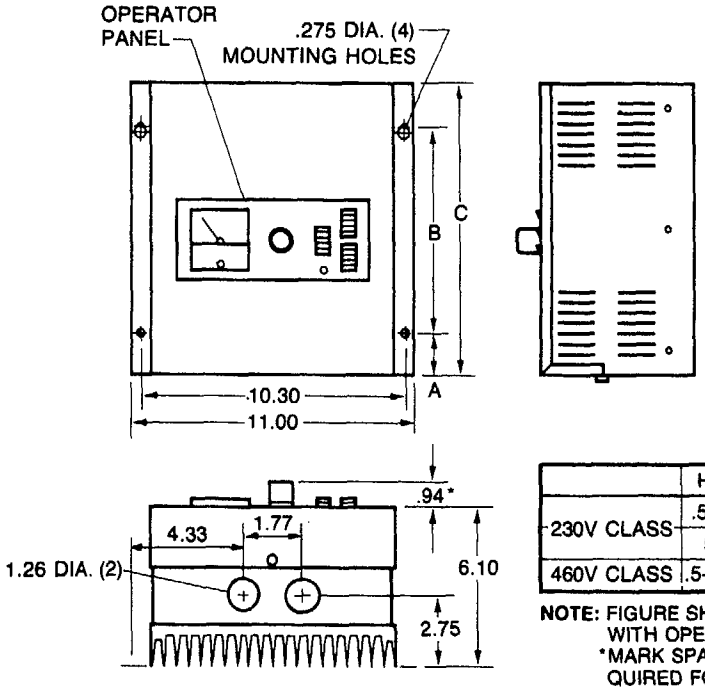


b) Connect #18 wire to TB 18 on main card.
c) Insert a female (5 Pin) connector to CN 10 male connector (middle, left) on main card.

Notes:
Use philips type #2 screwdriver so as not to round off screw heads.
Refer to Paragraph 7.3 for auto/manual control.

14. OUTLINE DRAWINGS

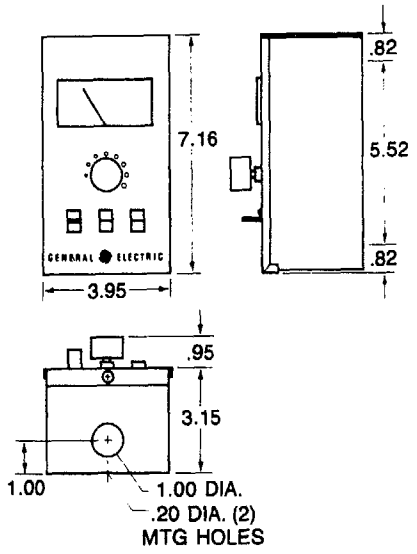
INVERTER UNIT



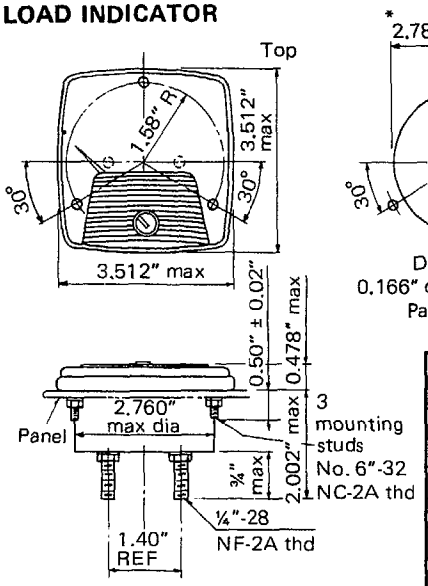
	HP	A	B	C
230V CLASS	.5-3	1.60	7.87	11.14
	5	1.24	11.80	13.90
460V CLASS	.5-10	1.24	11.80	13.90

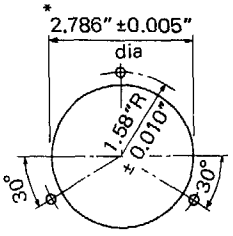
NOTE: FIGURE SHOWS "A" TYPE UNIT WITH OPERATOR PANEL.
*MARK SPACE IS NOT REQUIRED FOR "B" TYPE UNIT.

STANDARD CONTROL STATION 6VOC23A1



LOAD INDICATOR



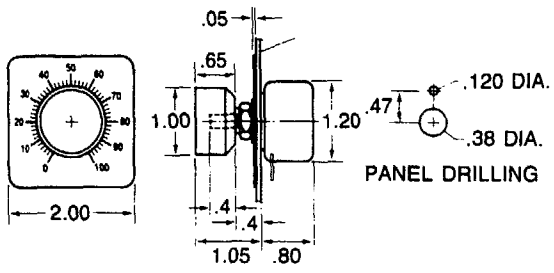


Drill 3 holes
0.166" dia (No. 19 drill)
Panel Drilling

Cat No.	HP		Max. Scale
	230 VAC	460 VAC	
6VLI20A1	–	0.5/1	3 AMP
6VLI21A1	0.5	2	5 AMP
6VLI22A1	1	3	8 AMP
6VLI23A1	2	–	10 AMP
6VLI24A1	3	5	15 AMP
6VLI25A1	–	7.5	20 AMP
6VLI26A1	5	10	30 AMP

(all dimensions in inches)

FREQUENCY ADJUST POTENTIOMETER 6VSP12A1 (2.5KΩ)



FREQUENCY INDICATOR – 6VFI 10A1

Scale : 0–120
Legend : FREQUENCY
Size : 3 1/2"

Outline same as load indicator

(all dimensions in inches)

15. AC LINE REACTORS

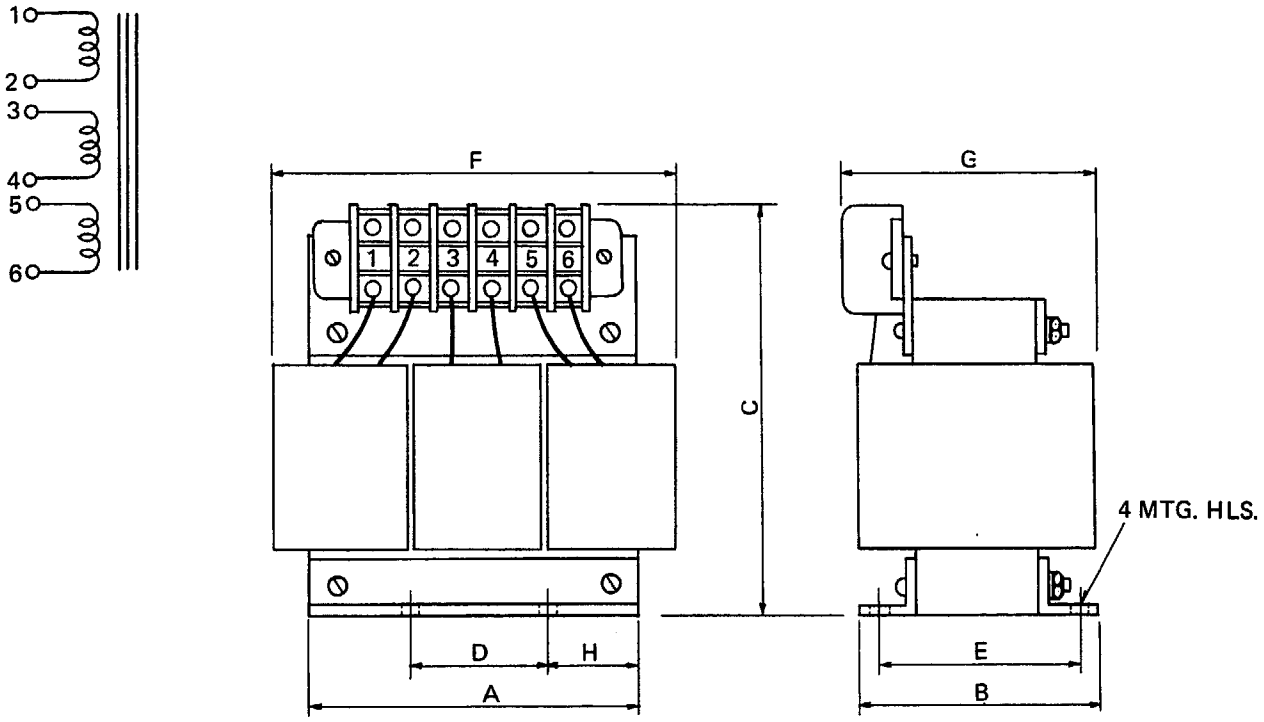
15.1

Table 12: Controllers/Reactors

Controller		AC Line
HP	Voltage	Reactor
1	230	104X220FB005
2	230	104X220FB006
3	230	104X220FB007
5	230	104X220FB008
1	460	104X220FB002
2-3	460	104X220FB004
5	460	104X220FB005
7.5-10	460	104X220FB006

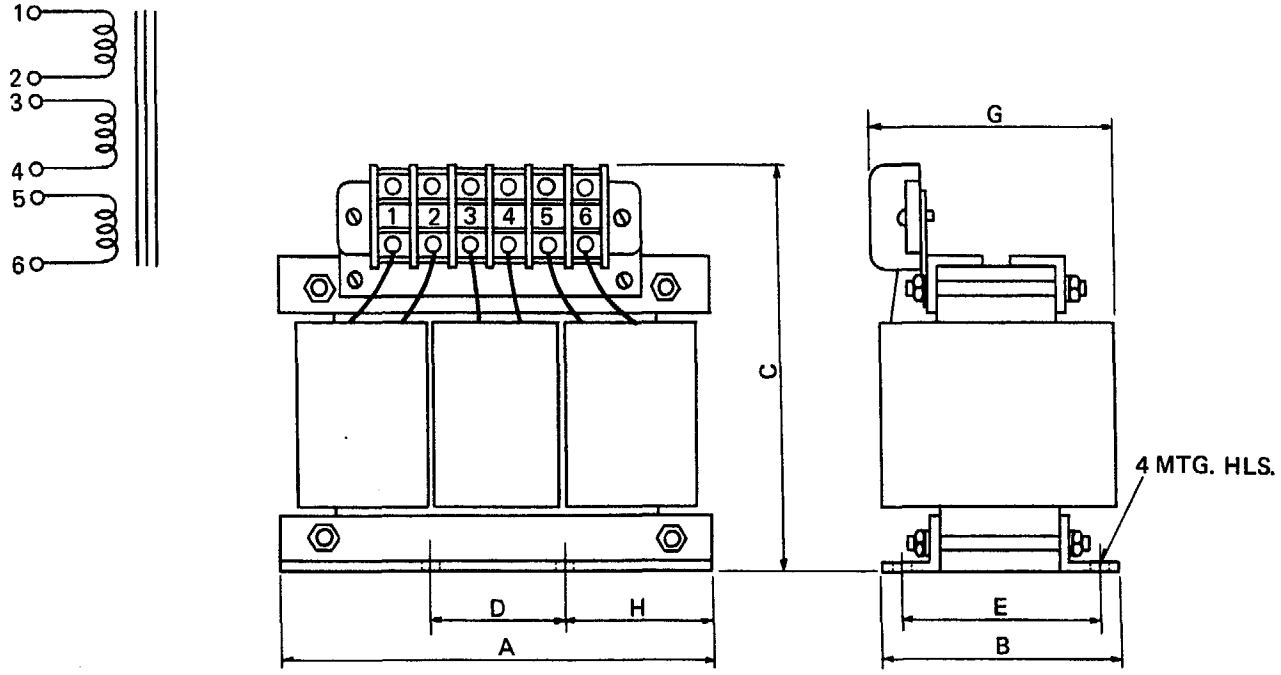
15.2 OUTLINE AND SCHEMATIC

104X220FB002, 004, 005, 008



See Table 13 for dimensions.

104X220FB006, 007



See Table 13 for dimensions

15.3

Table 13: Dimensions (In Inches)

GE #	A	B	C	D	E	F	G	H	Mtg. Holes
FB002	3.437	3.312	5.50	1.437	2.563	4.062	3.00	1.00	.25 X .50"
FB004	5.25	4.125	6.25	2.00	3.375	6.125	3.25	1.625	.25 X .37"
FB005	5.25	4.562	6.25	2.00	3.812	6.125	3.75	1.625	.25 X .37"
FB006	10.00	4.25	9.00	3.00	3.25	—	4.25	3.50	.437" DIA.
FB007	10.00	4.50	9.00	3.00	3.50	—	4.50	3.50	.437" DIA.
FB008	5.25	4.875	6.75	2.00	4.125	6.125	4.25	1.625	.25 X .37"

HOT LINE TELEPHONE NUMBER

The purpose of the following is to provide specific instructions to the AF-200E drive user regarding warranty administration and how to obtain assistance on out-of-warranty failures.

AF-200E DRIVE CONTROLLERS (1 to 10 HP)

The warranty covers all major parts of the controller such as main printed circuit boards, transistor modules, etc. but does not provide for replacement of fuses or the complete controller.

1. In the event of misoperation or failure during "in warranty period" refer to the instruction book to identify the defective part or subassembly.

2. When the defective part has been identified (or for assistance in identification) call:

GE COMPANY
GE DRIVE SYSTEMS
ERIE, PENNSYLVANIA
(814) 455-0521
(24-Hour Phone Service)

Before calling, list catalog numbers of the controller, motor, operator's station and any modification kits for ready reference. Obtain the serial number from the controller's nameplate.

AF-200E DRIVE MOTORS

AF-200E drive motor repairs are generally handled by General Electric Small Motor Service Centers or Apparatus Service shops. For specific instructions on your motor, call General Electric at the number listed above and furnish complete nameplate data.



GE Drive Systems

General Electric Company
GE Drive Systems
1100 Lawrence Parkway
Erie, PA 16531

GEK-85710B (3/90)